

MAE 4730
Lecture 37
11/26/2018

Today

- Review
- Dynamic balance

Reference configuration



$\underline{\underline{r}}_0, \underline{\underline{I}}_0$

↑
line drawn m object

Rotated Configuration

(same blob)



$\underline{\underline{I}}$

$$\underline{\underline{R}}: \underline{\underline{r}} = \underline{\underline{R}} \cdot \underline{\underline{r}}_0$$

$$\underline{\underline{I}} = \underline{\underline{R}} \underline{\underline{I}}_0 \underline{\underline{R}}^T$$

LMB: (simple)

$$\underline{\underline{F}}_{\text{ext}} = m \underline{\underline{a}}_G$$

$$\underline{\underline{\dot{v}}}_G = \underline{\underline{a}}_G$$

$$\underline{\underline{\dot{r}}}_G = \underline{\underline{v}}_G$$

KINEMATICS:

$$\underline{\underline{\dot{R}}} = \underline{\underline{\dot{R}}} \cdot \underline{\underline{R}}$$

$\mathcal{S}(\underline{\underline{\omega}}) \equiv \underline{\underline{\omega}} \times \equiv$ skew symmetric matrix

AMB:

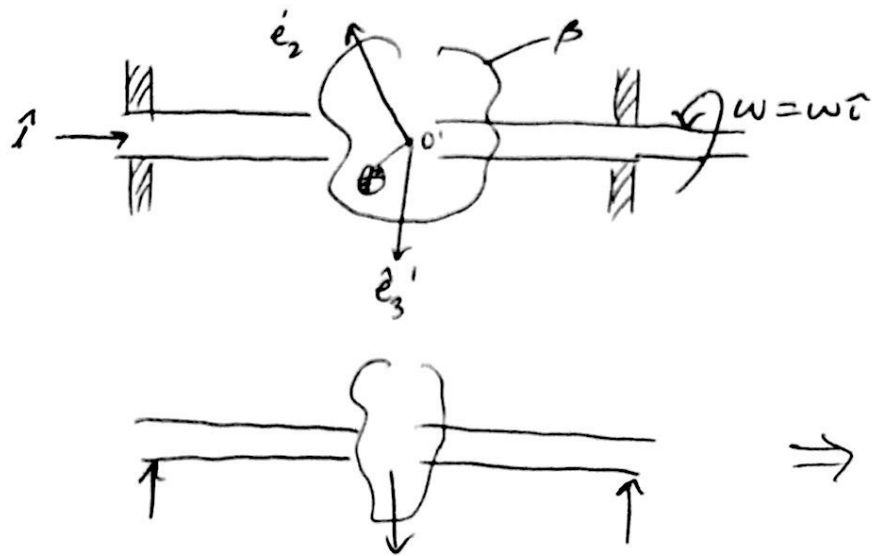
$$\underline{\underline{M}}_G^{\text{ext}} = \underline{\underline{H}}_G = \underline{\underline{I}} \cdot \underline{\underline{\omega}} + \underline{\underline{\omega}} \times (\underline{\underline{I}} \cdot \underline{\underline{\omega}})$$

$$\underline{\underline{\dot{\omega}}} = \underline{\underline{I}}^{-1} (\underline{\underline{M}}^{\text{ext}} - \underline{\underline{\omega}} \times (\underline{\underline{I}} \cdot \underline{\underline{\omega}}))$$

↑
from $\underline{\underline{r}}_G, \underline{\underline{R}}, \underline{\underline{\omega}}$

Famous Simple Solutions (pencil + paper)

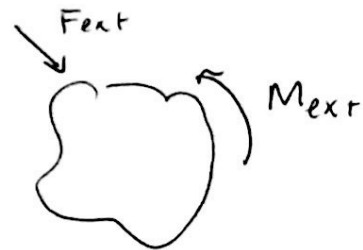
Constant ω (fixed axis rotation):



At time of instant

$$e_2 = e_2'$$

$$e_3 = e_3'$$



LMB:

$$\vec{F}_{ext} = m\vec{a}_G = m[\vec{\omega} \times (\vec{\omega} \times \vec{r}_{G/O}) + \cancel{\vec{a}_{O'/O}} + \cancel{\vec{\omega} \times \vec{r}}]$$

$F_{ext} \neq \text{constant}$ b/c $\dot{\vec{r}}_{G/O} \neq 0$

$F_{ext} = 0$ if G is on axis of rotation
($\vec{r}_{G/O} \parallel \vec{\omega}$)

"Static balance" $\iff G$ on axis of rotation

$\hookrightarrow$ can measure w/ static experiment

Assume static balance

- G on axis of rotation

$$\text{AMB/G: } \vec{M}^{ext} = \vec{H}_G = \underline{\underline{I}} \cdot \vec{\omega} + \vec{\omega} \times (\underline{\underline{I}} \cdot \vec{\omega})$$

$$M = \begin{cases} \vec{0} & \underline{\underline{I}} \cdot \vec{\omega} \parallel \vec{\omega} \\ \neq \vec{0} & \text{otherwise} \end{cases} \quad \vec{\omega} \text{ is eigenvector of } \underline{\underline{I}}$$

Dynamic balance

$\hat{z}$ is eigen-vector of $\underline{I}$

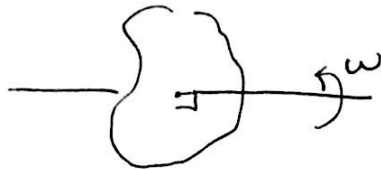
$\hat{z}$ is principal direction of $\underline{I}$

$$\begin{bmatrix} I_1 & 0 & 0 \\ 0 & \boxed{\text{Anything you want}} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \lambda \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

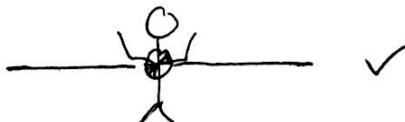
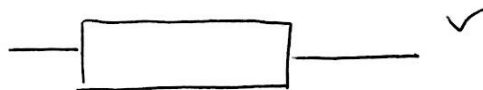
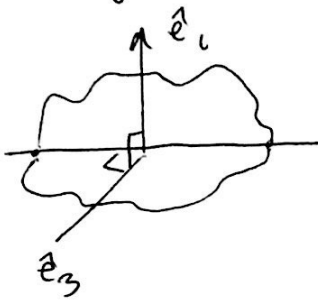
Examples)

Planar object w/ $\vec{\omega} \perp$ to plane will always be dynamically balanced

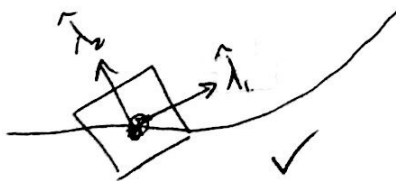
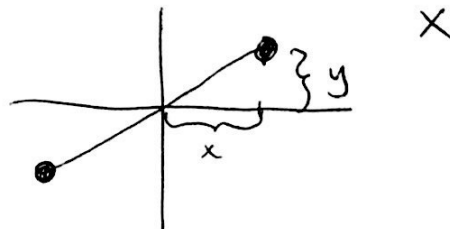
- planar
- prismatic



Planar object, axis in plane



$\checkmark$: dynamically balanced



Because 2 equal moment of inertia, so whole plane is a linear combination of eigenvectors